

Advanced (Carolina Reaper)**Open Ended Questions (FRQ)**

Q9. A water tank can be filled by two pipes. The amount of water filled by pipe A is given by $V_A = 2t^2 + 3t$, and the amount of water filled by pipe B is given by $V_B = t^2 - 2t$. If the total capacity of the tank is 1000 liters, find the time it takes to fill the tank

Q10. A cyclist is riding a bike with a speed given by $v = 3t^2 - 2t + 1$. If the cyclist also has to ride against a headwind given by $v_w = t + 1$, find the time when the cyclist's speed is equal to the wind speed

Chef's Tips

- Make sure students have a pencil, eraser, and calculator
- Remind students to show their work and explain their reasoning
- Emphasize the importance of checking their answers